ASIAN INSTITUTE OF MANAGEMENT

Machine Learning 3

Fundamentals of Machine Learning

Session 2

MSDS 2026

10 April 2026

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Fundamentals of Machine Learning

What is the goal of machine learning?

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Optimization vs. Generalization

OPTIMIZATION

- the process of adjusting a model to get the best performance possible on the training data

GENERALIZATION

- how well the trained model performs on data it has never seen before

Chollet, F., & Watson, M. (2026). *Deep Learning with Python Third Edition.* Manning.

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Optimization vs. Generalization

OPTIMIZATION

- the process of adjusting a model to get the best performance possible on the training data

GENERALIZATION

- how well the trained model performs on data it has never seen before

Can we control generalization?

Chollet, F., & Watson, M. (2026). *Deep Learning with Python Third Edition.* Manning.

What happens if we fit a model to its training data too well?

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Overfitting and underfitting

OVERFITTING

- the model learns patterns that are specific to the training data but that are misleading or irrelevant when it comes to new data

Chollet, F., & Watson, M. (2026). *Deep Learning with Python Third Edition.* Manning.
Goodfellow, I., Bengio, Y., Courville, A., & Bengio, Y. (2016). *Deep learning* (Vol. 1, No. 2, pp. 1-800). Cambridge: MIT press.

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Overfitting and underfitting

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What happens to the validation metrics?

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Overfitting and underfitting

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UNDERFITTING

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Overfitting and underfitting

OVERFITTING

- the model learns patterns that are specific to the training data but that are misleading or irrelevant when it comes to new data

UNDERFITTING

- the model is not able to obtain a sufficiently low error value on the training and validation sets

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Overfitting and underfitting

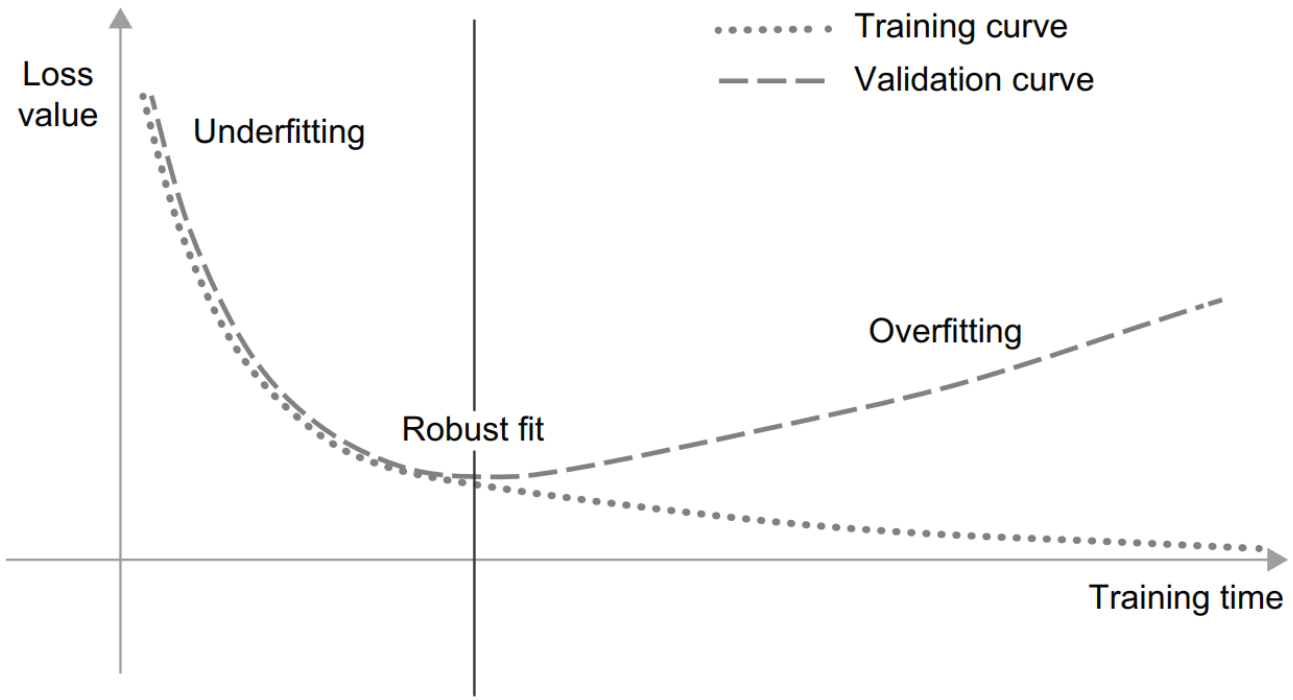

Figure 5.1 Canonical overfitting behavior

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How should the generalization power of a model be reliably measured?

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Evaluating ML models: DATA SPLITTING

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Evaluating ML models: DATA SPLITTING

Into how many sets should the available data be split?

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Evaluating ML models: DATA SPLITTING

Evaluation protocol

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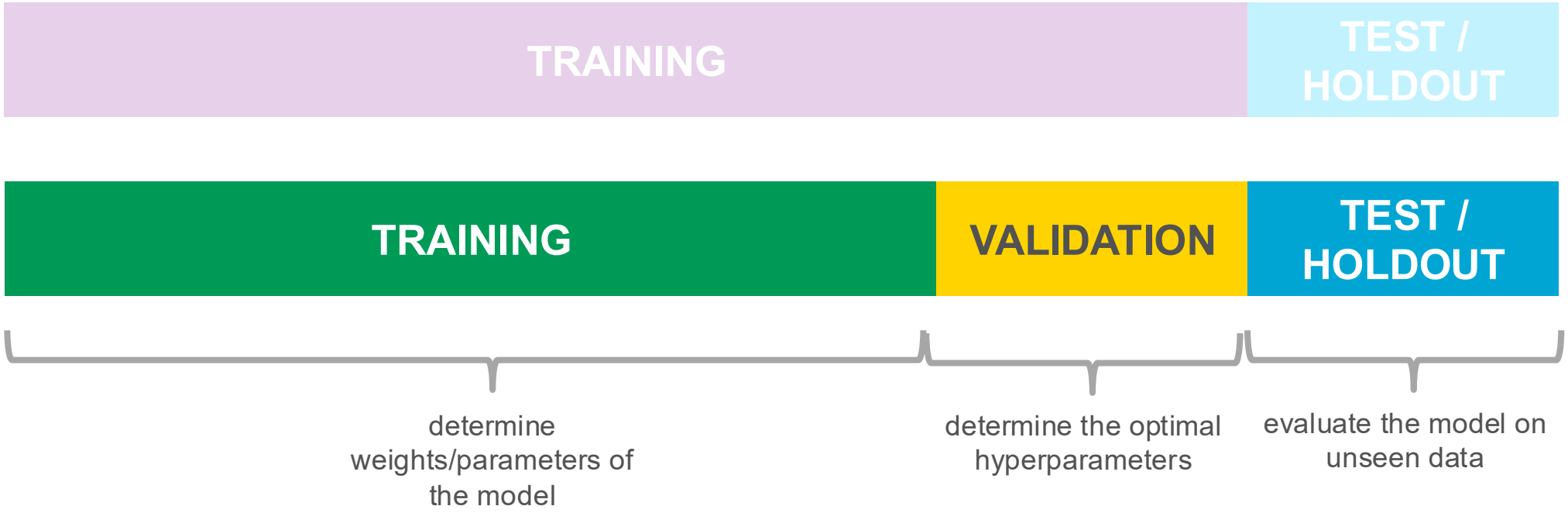
Evaluating ML models: DATA SPLITTING

Evaluation protocol



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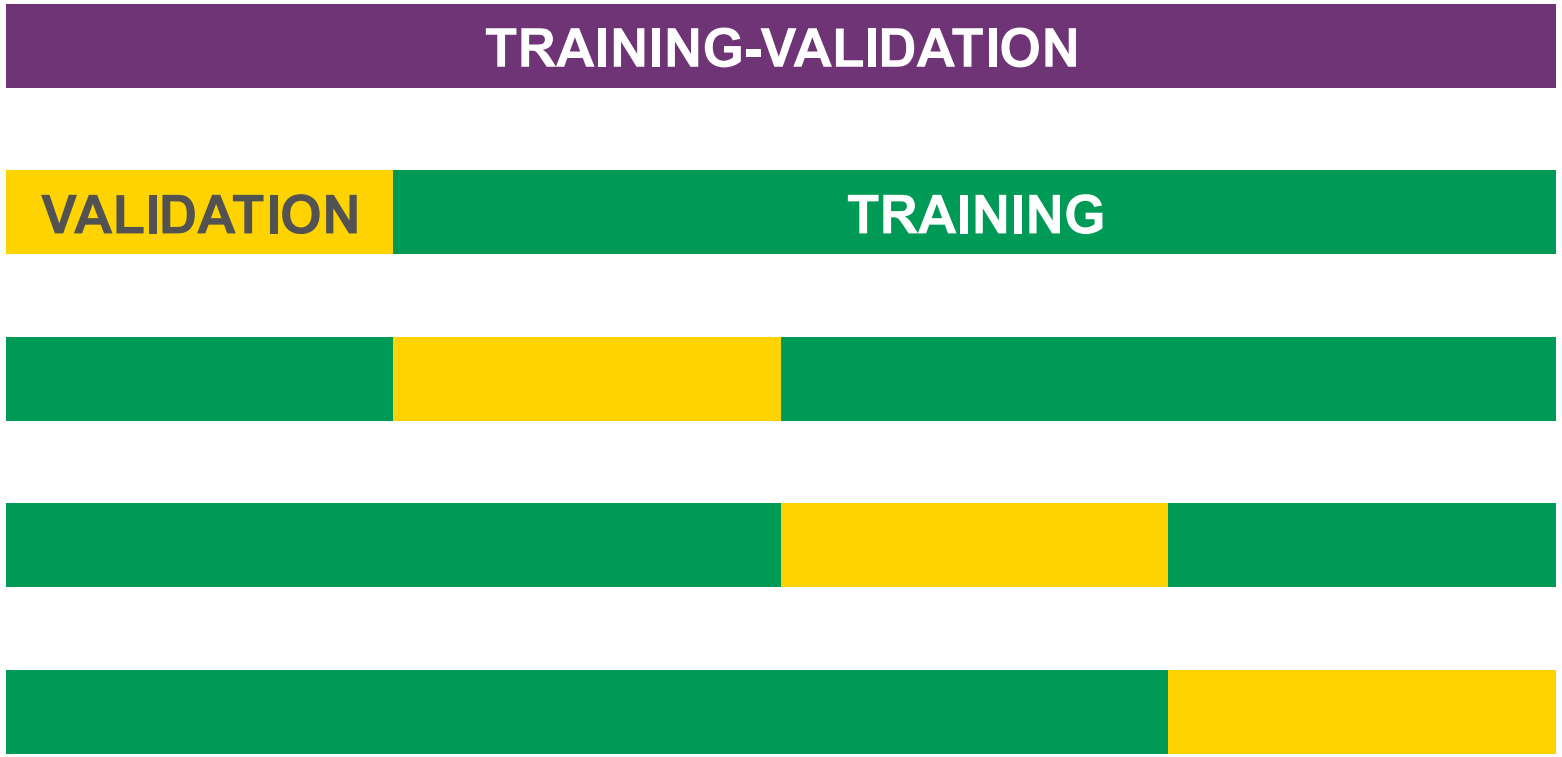
Evaluating ML models: DATA SPLITTING

Evaluation protocol



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Evaluating ML models: DATA SPLITTING

Evaluation protocol

- Maintaining a test set
- Doing K-fold cross-validation
- Doing iterated K-fold validation

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Evaluating ML models: DATA SPLITTING

Important considerations

- Data representativeness
- The arrow of time
- Redundancy of data

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Evaluating ML models: BEATING A COMMON-SENSE BASELINE

Evaluating ML models: BEATING A COMMON-SENSE BASELINE

What are examples of common-sense baselines?

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Evaluating ML models: BEATING A COMMON-SENSE BASELINE

When should you pick a trivial baseline?

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Evaluating ML models: BEATING A COMMON-SENSE BASELINE

What does it indicate if you can’t beat a trivial solution? What should you do?

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Evaluating ML models: BEATING A COMMON-SENSE BASELINE

Important points

- **Before** you start working with a dataset, you should always pick a trivial baseline that you'll try to beat.

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Evaluating ML models: BEATING A COMMON-SENSE BASELINE

Important points

- **Before** you start working with a dataset, you should always pick a trivial baseline that you'll try to beat.
- If you can't beat a trivial solution, your model is worthless. Go back to the drawing board.
  - Are you using the wrong model?
  - Can the problem be really approached with machine learning?

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Optimization vs. Generalization

Overfitting and Underfitting

Evaluating ML Models

- Data splitting
- Beating a common-sense baseline

Universal Workflow of Machine Learning

Universal Workflow of ML

Define the task

- Frame the problem
- Collect a dataset
- Understand your data
- Choose a success measure

Develop a model

- Prepare the data
- Choose an evaluation protocol
- Beat a baseline
- Develop a model that overfits
- Regularize and tune the model

Deploy the model

- Explain work to stakeholders and set expectations
- Ship an inference model
- Monitor the model in the wild
- Maintaining the model

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Universal Workflow of ML

Define the task

- Frame the problem
- Collect a dataset
- Understand your data
- Choose a success measure

- Data available
  - Inputs
  - Outputs
- Broad type of ML task

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Universal Workflow of ML

Define the task

- Frame the problem
- **Collect a dataset**
- Understand your data
- Choose a success measure

Points to consider

- Properties of the data
- Data annotation process
- Nonrepresentative data

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Universal Workflow of ML

Define the task

- Frame the problem
- Collect a dataset
- **Understand your data**
- Choose a success measure

- Explore and visualize the data

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Universal Workflow of ML

Define the task

- Frame the problem
- Collect a dataset
- Understand your data
- **Choose a success measure**

- Define what you mean by success
- The measure should be aligned with your higher-level goals

Universal Workflow of ML

Define the task

- Frame the problem
- Collect a dataset
- Understand your data
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Universal Workflow of ML

Develop a model

- **Prepare the data**
- Choose an evaluation protocol
- Beat a baseline
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- Data vectorization
- Value normalization
- Handling missing values

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Universal Workflow of ML

Develop a model

- Prepare the data
- **Choose an evaluation protocol**
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Common Evaluation Protocols

- Maintaining a test set
- Doing K-fold cross-validation
- Doing iterated K-fold validation

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Universal Workflow of ML

Develop a model

- Prepare the data
- Choose an evaluation protocol
- **Beat a baseline**
- Develop a model that overfits
- Regularize and tune the model

Points to consider

- Feature engineering
- Selecting the correct architecture priors
  - Type of model architecture
- Selecting a good enough training configuration
  - Loss function, batch size, learning rate

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Universal Workflow of ML

Develop a model

- Prepare the data
- Choose an evaluation protocol
- Beat a baseline
- **Develop a model that overfits**
- Regularize and tune the model

- Add layers
- Make the layers bigger
- Train for more epochs

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Universal Workflow of ML

Develop a model

- Prepare the data
- Choose an evaluation protocol
- Beat a baseline
- Develop a model that overfits
- **Regularize and tune the model**

- Try different architectures
- Add dropout
- Add L1 or L2 regularization
- Try different hyperparameters
- Iterate on data curation or feature engineering

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Universal Workflow of ML

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Introduction to Keras

KERAS

Our DL Framework of Choice

TensorFlow

PyTorch

Google JAX

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Exercise 2

Reminders

Reminders

Next Session

Session 3 and 4 (April 17, 2026) – Convolutional Neural Network

Deliverables

Due this week

- Post-Class Reflection 1 (due EOD April 10, 2026)

Due next week

- Exercise 1 (due EOD April 11, 2026)
- Exercise 2 (due EOD April 15, 2026)

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